GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Research Report

Mini research report applying the research methodology (Introduction, Method, Results, Discussion)

Student Name	Kothari Tarj Mehulbhai	Enrollment No.	260205020001
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Activity	ALA 5	Marks	10

Comparative Search Efficiency of a Randomized Skip List, a Deterministic Red-Black Tree and a Self-Adjusting Splay Tree under Different Insertion Orders and Access Patterns

Abstract

Background: Ordered search structures can be balanced deterministically (red-black trees), adapt to the access sequence (splay trees) or use randomization (skip lists). Their relative search efficiency under different insertion orders and access patterns is rarely tested with replicated experiments and statistical inference. **Method:** A factorial computer experiment (3 structures × 2 insertion orders × 2 query patterns, 30 independent replications each, n = 5000 keys, 3000 queries per run) measured the mean number of nodes inspected per search. Hypotheses were tested with Welch t-tests, one-way ANOVA with Tukey HSD,anequivalencetestandachi-squaregoodness-of-fittest.**Results:**Underuniformqueriesthe red-black tree needed 11.62 nodes per search, the splay tree 21.26 and the skip list 24.31 (all differences $p < 0.001$). With skewed queries the splay tree (10.33) beat the red-black tree (11.59). Sorted insertion increased splay-tree cost significantly, whereas no significant effect was found for the skip list. **Conclusion:** The skip list's measured cost matches theory ($2 \cdot \log_2 n \approx 24.6$), but in a sequential setting it inspects about twice as many nodes as a red-black tree; its advantages lie in simplicity and insensitivity to input order.

Keywords:skiplist;red-blacktree;splaytree;randomizedalgorithms;searchcost;replicatedexperiment; hypothesis testing.

1. Introduction

Background

Searching an ordered collection is a basic operation in databases, compilers and operating systems. A plain binary search tree can degrade to a linked list, so three families of remedies exist: **deterministic balancing** (Bayer and McCreight, 1972; Guibas and Sedgewick, 1978), **self-adjusting** trees (Sleator and Tarjan,1985)and**randomizedbalancing**(AragonandSeidel,1989;Pugh,1990).Theorydescribeseach of them, but the guarantees are of different types (worst-case, amortized, expected), so a measured comparison is needed to see how they behave together.

Problem statement, objectives and hypotheses

Problem: It is not clear how much search efficiency is gained or lost by choosing a randomized skip list instead of a deterministic red-black tree or a self-adjusting splay tree, and whether this depends on the insertion order and on the query pattern.

- **O1** Measure the average search cost of the three structures under uniform queries.
- **O2** Test whether the splay tree outperforms the red-black tree under skewed queries.
- **O3** Test whether the skip list is insensitive to insertion order, in contrast to the splay tree.
- **O4** Compare the measured skip-list cost with theory and check the distribution of random levels.

ID	Hypothesis (null form)
H1	Skip list and red-black tree have equal mean search cost (uniform queries, random insertion).
H2	Splaytreeandred-blacktreehaveequalmeansearchcostunderskewedqueries(alternative:splayis lower).
H3	Skip-list search cost is the same for sorted and random insertion.
H4	Splay-tree search cost is the same for sorted and random insertion.
H5	Skip-list levels follow Geometric(0.5).
H6	All three structures have equal mean search cost (uniform queries).

2. Literature Review and Research Gap

The structured review (ALA 1) covered nine articles. Pugh (1990) introduced skip lists and reported that they are simple and competitive with balanced trees; Papadakis, Munro and Poblete (1992) derived exact expected search costs and showed that Pugh's upper bound is fairly tight; Munro, Papadakis and Sedgewick (1992) gave deterministic variants with worst-case guarantees; Sleator and Tarjan (1985) proved amortized logarithmic cost for splay trees; Herlihy et al. (2007) showed that skip lists are practical in concurrent settings; and Georges et al. (2007) argued that performance claims need replicated, statistically rigorous evaluation. **Gap:** the reviewed works are mostly analytical or report few un-replicated timings, and rarely combine several insertion orders and access patterns with replicated runs and formal hypothesis tests.

3. Methodology

Research design

A **quantitative, experimental, applied** design: a full factorial computer experiment with independent replications. The three structures were implemented in Python and verified by randomized tests against a reference set (12,000 random insert/delete operations plus search checks per structure) before measurement.

Element	Specification
Factors	Structure(skiplistwithp=0.5,red-blacktree,splaytree);insertionorder(random,sorted); query pattern (uniform; skewed = 90% of queries to a hot set of 1% of the keys)
Dependent variable	Mean nodes inspected per successful search (counted in code); secondary: μs per search
Populationand sampling	Keys: simple random sample of 5000 integers from 1–999,999 per replication. Queries: 3000perrundrawnatrandomfromthestoredkeys(uniform)orfromahotset(skewed). Replications: 30 independent seeds.
Data collection	AutomaticmeasurementharnesswritingoneCSVrowperrun(360rows);nomissing values; deterministic node counts for a given seed (ALA 4).

Element	Specification
Analysis plan	Descriptivestatisticswith95%confidenceintervals;Shapiro–WilkandLevenetests;Welch t-tests (H1–H4) with Holm correction; Mann–Whitney as robustness check; TOST equivalence test (H3); one-way ANOVA + Tukey HSD (H6); chi-square (H5). α= 0.05.
Ethics	Nohumanparticipants.Code,seedsandrawdataareprovidedforverification;allresults are reported, including non-significant ones.

4. Results

Descriptive statistics

Structure	Insertion	Queries	n	Mean	SD	95% CI of mean	µs/search
SkipList	random	uniform	30	24.31	1.69	[23.68, 24.95]	1.98
SkipList	random	skewed	30	24.38	1.9	[23.67, 25.09]	1.98
SkipList	sorted	uniform	30	24.73	1.8	[24.06, 25.41]	2.17
SkipList	sorted	skewed	30	24.78	1.86	[24.08, 25.47]	1.89
RedBlack	random	uniform	30	11.62	0.04	[11.61, 11.64]	0.77
RedBlack	random	skewed	30	11.59	0.23	[11.51, 11.68]	0.67
RedBlack	sorted	uniform	30	11.89	0.04	[11.88, 11.91]	0.75
RedBlack	sorted	skewed	30	11.89	0.24	[11.80, 11.98]	0.67
Splay	random	uniform	30	21.26	0.09	[21.22, 21.29]	2.45
Splay	random	skewed	30	10.33	0.09	[10.30, 10.36]	1.31
Splay	sorted	uniform	30	23.59	0.11	[23.55, 23.63]	2.73
Splay	sorted	skewed	30	12.87	0.11	[12.83, 12.91]	1.61

Mean nodes inspected per search over 30 replications.

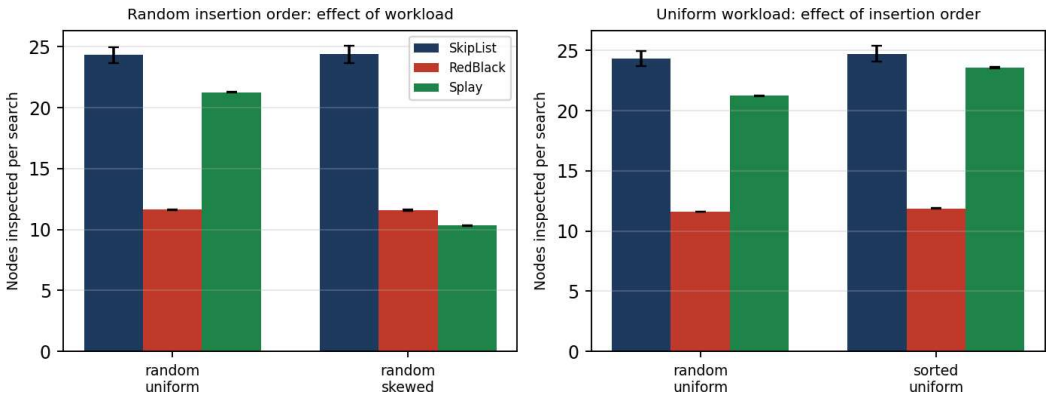


Figure 1: Mean search cost with 95% confidence intervals for all conditions.

Hypothesis tests

ID	Comparison	Mean diff.	95% CI of diff.	t (Welch)	Holm p	d
H1	Skiplist–red-black(uniform, random)	12.69	[12.06, 13.32]	41.15	<0.001	10.62
H2	Splay –red-black (skewed, random)	-1.26	[-1.35, -1.17]	-28.02	<0.001	-7.23
H3	Skip list: sorted –random	0.42	[-0.48, 1.32]	0.93	0.358	0.24
H4	Splay: sorted –random	2.34	[2.29, 2.39]	94.06	<0.001	24.29

Mean diff. in nodes per search. H2 is one-tailed. d = Cohen's d.

- **H1 rejected:** the skip list inspected 12.69 more nodes per search than the red-black tree (a factor of 2.1).
- **H2 rejected:** under skewed queries the splay tree inspected 1.26 fewer nodes than the red-black tree (10.33 vs 11.59).
- **H3 not rejected** (p = 0.358); the 95% CI of the difference [-0.48, 1.32] is not within ±1 node, so equivalence (TOST p = 0.101) cannot be claimed.
- **H4 rejected:** sorted insertion raised the splay-tree cost by 2.34 nodes (23.59 vs 21.26).
- **H6 rejected:** one-way ANOVA F(2, 87) = 1379.8, p < 0.001, η^2 = 0.969; Tukey HSD showed that all three pairs differ (p < 0.001).
- **H5 not rejected:** $\chi^2(10) = 6.86$, p = 0.739: the random levels are consistent with Geometric(0.5).

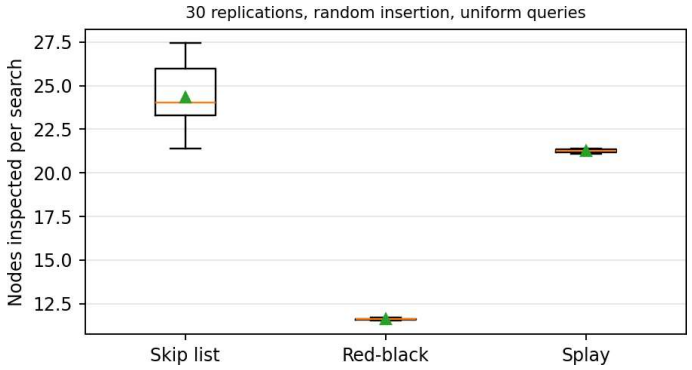


Figure2:Distributionofthereplicationmeans(uniformqueries,randominsertion).Theorderingred-black<splay<skiplist was found in every replication.

5. Discussion

Interpretation

- **Agreement with theory.** The measured skip-list cost (24.31, 95% CI [23.68, 24.95]) is close to the expectedvalue2·$\log_2$(5000)≈24.6(Papadakisetal.,1992;Pugh,1990),andthered-blacktree(11.62) is close to $\log_2$(5000) ≈12.3. This supports the correctness of the implementations and the measurement method.
- **Price of randomization.** In a sequential setting the skip list needs about twice as many node inspections as a red-black tree, because it walks along several levels and must test the next node at each level before dropping down. The wall-clock measure (secondary) gave the same ordering of the red-black tree as the fastest.
- **Adaptivity.** The splay tree is worse than the red-black tree for uniform queries but better for skewed queries, as predicted by Sleator and Tarjan (1985): frequently used keys move to the top.
- **Insertionorder.**Theshapeofthesplaytreedependsonitshistory(sortedinsertiongave+2.3nodes), whereas for the skip list no significant effect was found. Because the CI of the difference is wide (about ±1), a small order effect cannot be excluded.
- **Practicalmeaning.**Ifguaranteedperformanceisthepriority,thered-blacktreeisthebestofthethree; if access is highly skewed, a splay tree may be preferable; the skip list is attractive for its simple code and,asshowninthe literature(Herlihyetal.,2007),forconcurrentuse,butnotforminimumsequential search cost.

Threats to validity

Type	Threat	Mitigation / remaining risk
Construct	‘Nodesinspected’dependsoncountingrules that differ slightly between structures (e.g. skip-list comparisons at every level; two comparisons per node in splaying).	Same rule applied consistently in each structure;orderingconfirmedbywall-clocktime; absolute ratios should be interpreted cautiously.
Internal	Implementation quality could affect results.	Allstructurestestedagainstareferenceset; seeds fixed; same machine and code base.
External	Singlesizen=5000,Python,integerkeys, single thread, one skew level, successful searches only.	Resultsshouldnotbegeneralisedtoverylarge data, other languages, caches or concurrency (not tested).
Statistical conclusion	Verysmallvariance(red-black)makestiny differences significant; multiple tests.	EffectsizesandCIsreported;Holmcorrection; non-parametric check; equivalence test for H3.

6. Conclusion and Future Work

Under the conditions studied, (i) the red-black tree had the lowest and most stable search cost for uniform queries;(ii)thesplaytreeoutperformeditforskewedqueries;(iii)theskiplist'scostmatcheditstheoretical value but was about twice that of the red-black tree, and was not significantly affected by insertion order, while the splay tree was; and (iv) the skip list's random levels followed the intended geometric distribution. **Futurework:**varyn,addinsertanddeleteoperations,includethedeterministicskiplistandtreaps,repeat in a compiled language with cache effects, test concurrent access, and complement the experiment with the planned developer survey and interviews (ALA 4).

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Appendix: Reproducing the Study

Files supplied with the activities: `RM_measurement_harness.py` (experiment) and `RM_experiment_data.csv`(360observations). Running the harness regenerates the data because all random numbers are seeded (node counts are exactly reproducible; timings vary by machine). The data structure implementations are those of the Advanced Data Structure activities (skip list, red-black tree, splay tree).